Tristan Tan

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Education

University of California, Berkeley — Berkeley, CA

Data Science & Economics; GPA: 3.88 — *Aug 2025 – May 2029*

- Relevant Coursework: Microeconomics, Probability & Statistics for Business, Foundations of Data Science, Computational Structures in Data Science, Linear Algebra

Technical Skills

Languages: Python, SQL, R, JavaScript, TypeScript, HTML/CSS
Data Science & ML: Pandas, NumPy, SciPy, scikit-learn, XGBoost, PyTorch, Sentence Transformers, OpenCV
Visualization: Tableau, Matplotlib
Frameworks & Tools: React, Next.js, FastAPI, PostgreSQL, Supabase, MongoDB, Git, Vercel, OAuth 2.0, REST APIs

Experience

Undergraduate Research Assistant — Sep 2026 – Present

Archaeological Research Facility (ARF), UC Berkeley — *Berkeley, CA*

- Developed an interactive 3D annotation platform using Next.js, TypeScript, and React Three Fiber, integrating 314 scholarly annotations with 607 spatial regions across 6 photogrammetric Egyptian coffin models
- Implemented mesh raycasting and bidirectional spatial selection, linking 3D hieroglyphic inscriptions to scholarly translations and enabling direct navigation between annotated surfaces and textual records
- Built a structured annotation data pipeline integrating existing scholarly datasets with interactive 3D visualization, reducing reliance on manual XML lookup and supporting scalable digital humanities research

Data Research Intern — Sep 2026 – Present

Ten Strands — *Berkeley, CA*

- Built an NLP retrieval system combining BM25, dense transformer embeddings, and a CrossEncoder reranker to surface sustainability evidence across 10,000+ California K–12 district webpages and policy documents
- Benchmarked lexical and neural retrieval on human-validated evidence using Recall@K and MRR, improving Recall@5 by 35% and reducing manual source-discovery time by 76%

Data Science Intern — May 2026 – Aug 2026

IDX Exchange — *Remote*

- Engineered an end-to-end Python/scikit-learn ML pipeline for automated residential property valuation on 297K+ California real estate transactions across 27 months, covering data cleaning, feature engineering, and model training
- Optimized Linear Regression, Random Forest, and XGBoost models with chronological backtesting and hyperparameter tuning across 11K+ property sales, achieving 0.92 R^2, 6% median prediction error, and 34% lower MAE than the linear baseline

Undergraduate Research Apprentice — Aug 2025 – May 2026

Lucarelli Photogrammetry Lab, UC Berkeley — *Berkeley, CA*

- Built an automated Python photogrammetry pipeline for archaeological artifacts using the Agisoft Metashape API, processing 1,000+ images into 3D reconstructions and cutting manual processing time by 40% (∼30 to ∼18 min)
- Built Ridge regression models on SfM and image-overlap graph features to predict 3D reconstruction quality, achieving 13.7% lower geometric RMSE than a reprojection-error baseline under leakage-aware CV

Projects

Spotify Analytics Platform | *Python, FastAPI, SQLAlchemy, PostgreSQL, Spotify Web API, OAuth 2.0*

- Built a FastAPI and PostgreSQL backend that authenticates users through Spotify OAuth 2.0 and ingests their top tracks, artists, and listening history into a normalized relational schema
- Found that Spotify's audio-features endpoint is deprecated and its artist-genre data returns empty in practice, and redesigned the recommender around recency-weighted artist affinity instead

Quantitative ML Stock Analytics Dashboard | *Python, Pandas, scikit-learn, Streamlit, yfinance*

- Built an end-to-end financial analytics platform that ingests live market data, computes technical indicators (RSI, MACD, Bollinger Bands), and visualizes market trends in an interactive Streamlit dashboard
- Developed and evaluated Support Vector Regression (RBF) and Random Forest models using time-series-aware train/test splits, feature scaling, and out-of-sample testing on up to 365 days of historical market data